

Evaluation

User Evaluation

Four participants evaluated Docent in a controlled simulated-gallery environment. The study took place in a booked room at Queen Mary University of London using printed reproductions of the artworks available to the guide. Participants were free to inspect the paintings, move between them, ask questions and otherwise interact with Docent as they would with a human gallery guide. The system interface was deliberately visually de-emphasised, with the display brightness reduced so that participants could not rely on captions or dialogue transcripts and interaction remained primarily conversational. This was particularly important during conversational breakdowns: participants responded to what they had actually heard rather than being able to inspect the system's transcription and anticipate that they had been misrecognised. A fuller description of the procedure and study conditions is provided in the Study Protocol.

Evaluation combined dialogue transcripts, technical telemetry, post-task questionnaire responses and semi-structured interviews. It was conducted in three stages: an adapted PARADISE analysis quantified local conversational success, dialogue costs and user satisfaction; a conversational-principles analysis assessed the interaction behaviours targeted by the system design; and finally, observed failures were traced back to their likely technical or architectural origins.

PARADISE Evaluation

PARADISE (PARAdigm for Dialogue System Evaluation) is a framework for evaluating spoken dialogue agents by modelling performance as a combination of task success and dialogue costs, with user satisfaction providing the external criterion against which these factors are assessed. The framework was developed to permit comparison between dialogue strategies while separating what a user achieves from the particular dialogue behaviour used to achieve it. In its original form, PARADISE uses multivariate regression to estimate the relative contribution of success and cost variables to user satisfaction (Walker et al., 1998).

Unlike the bounded information-retrieval tasks considered in the original PARADISE studies, Docent supports exploratory conversation without a single terminal task. Task success was therefore operationalised at the level of the local conversational objective (LCO): the immediate objective that the participant attempted to accomplish through one or more dialogue turns.

Local conversational objectives

An LCO began when the participant expressed an information or conversational objective and ended when that objective was satisfied, abandoned, or superseded by another objective. Clarifications and reformulations of the same request were retained within the same LCO rather than counted as additional objectives. Purely social openings, acknowledgements and closings were excluded.

Participant	LCOs
P-01	55
P-02	22
P-03	14
P-04	19
Total	110

Each LCO was rated using a three-level task-success scale: 2 = objective successfully satisfied; 1 = objective materially addressed but only partially satisfied; 0 = objective not meaningfully satisfied.

Participant	Successful	Partial	Failed	Points / Maximum	LGS
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P-01	41	13	1	95 / 110	0.864
P-02	14	8	0	36 / 44	0.818
P-03	10	4	0	24 / 28	0.857
P-04	16	3	0	35 / 38	0.921
Overall	81	28	1	190 / 220	0.864

The system therefore achieved 86.4% of the maximum local-goal-success score, indicating that most participant objectives were ultimately satisfied.

Dialogue costs

PARADISE treats task success separately from the interaction cost incurred in achieving it. Dialogue costs were therefore derived independently from the transcripts and telemetry. These included ASR errors, user repairs, failures of contextual continuity, additional turns required to resolve an objective, response latency, abandoned objectives, interruption/turn-fragmentation events, and audio-buffer underruns.

Where the transcript included both an ASR transcription and an Actual utterance, the latter was treated as the spoken ground truth. This allowed ASR failure to be distinguished from semantic success: a recognition error could occur while the system nevertheless inferred the intended request and answered correctly.

Participant	ASR error rate	Repair rate	Context failure	Turns to resolution	Mean response latency
P-01	25.5%	9.1%	0.0%	1.13	2.07 s
P-02	9.1%	0.0%	0.0%	1.00	4.33 s
P-03	28.6%	28.6%	21.4%	1.43	13.08 s
P-04	21.1%	0.0%	0.0%	1.00	7.50 s

The separation of ASR error from dialogue success was particularly informative. P-01 and P-04 contained several inaccurate transcriptions that nevertheless resulted in an appropriate response, whereas missed utterances in P-03 propagated into incorrect conversational context and subsequent repair.

User satisfaction

User satisfaction was calculated from the 15 items in the post-task questionnaire. Items 15–18 described negative experiences and were therefore reverse-coded before averaging. The pre-task questionnaire was used only for participant background and was not incorporated into the evaluation score.

Participant	Post-task total	Maximum	Mean satisfaction / 7
P-01	77	105	5.13
P-02	78	105	5.20
P-03	71	105	4.73
P-04	98	105	6.53

The quantitative measures were broadly consistent with the post-task interviews. P-04 produced both the highest Local Goal Success score and the highest user-satisfaction score. By contrast, P-03 achieved a comparable level of task success to P-01 but exhibited considerably greater latency, repair and contextual failure, corresponding with the lowest satisfaction score and explicit complaints regarding response delay and topic transitions.

Normalisation and limitation of the PARADISE model

PARADISE normally standardises performance measures and uses multivariate regression to estimate their contribution to user satisfaction. The measures in this study were similarly normalised to permit descriptive comparison. However, the study contained only four independent participant-level satisfaction observations. Estimation of a regression-derived performance function from this sample would therefore be statistically unreliable and highly susceptible to overfitting.

The evaluation consequently adopts PARADISE as a descriptive evaluation framework rather than attempting to derive generalisable performance weights. This retains its distinction between task success, interaction cost and user satisfaction while avoiding unsupported statistical inference.

A particularly clear result is provided by P-01 and P-03. Their Local Goal Success scores were almost identical (0.864 and 0.857 respectively), yet P-03 exhibited substantially greater repair, contextual failure, turns to resolution and latency and reported lower satisfaction. This demonstrates that successful completion of conversational objectives alone did not adequately characterise the quality of the spoken interaction.

Conversational-Principles Evaluation

PARADISE primarily evaluates whether objectives were achieved and at what interaction cost. A second analysis therefore considered whether the dialogues exhibited the conversational behaviours targeted by the design of Docent. These principles were derived from the interaction requirements established earlier in the project: naturalistic turn-taking, floor management, grounding and repair, contextual continuity, pragmatic interpretation, dialogue-driven navigation, and appropriately scoped information delivery.

Each principle was evaluated only when the dialogue provided a genuine opportunity for that behaviour to be demonstrated. Opportunities were rated as 2 = successfully handled; 1 = partially handled; 0 = failed. The score for each principle represents the points obtained divided by the maximum available points. Opportunity counts are retained because some behaviours, particularly interruption and floor management, occurred considerably less frequently than ordinary contextual follow-ups or information requests.

Evaluation criteria

Principle	Evaluation criterion
Turn-taking	Appropriate detection of turn completion without premature finalisation or loss of the user's contribution
Floor management / interruption	Successful handling of interruptions, continuation, overlap and changes in floor ownership
Grounding and repair	Recognition and recovery from misunderstanding, ambiguity or correction
Contextual continuity	Maintenance and appropriate updating of artwork, topic and referential context
Pragmatic interpretation	Resolution of indirect, elliptical or context-dependent user meaning
Dialogue-driven navigation	Movement between artworks or conversational directions through natural dialogue rather than explicit interface control
Information delivery	Provision of an appropriate amount of information for the request without material under- or over-answering

Conversational performance

Principle	Overall opportunities	Points / Maximum	Overall score
Contextual continuity	65	122 / 130	93.8%
Information delivery	110	199 / 220	90.5%
Pragmatic interpretation	42	74 / 84	88.1%
Dialogue-driven navigation	31	54 / 62	87.1%
Grounding and repair	21	31 / 42	73.8%
Turn-taking	12	14 / 24	58.3%
Floor management / interruption	7	7 / 14	50.0%

The results indicate a distinction between the system's semantic conversational capabilities and the mechanics of real-time spoken interaction. Contextual continuity, pragmatic interpretation, dialogue-driven navigation and information delivery all achieved scores above 85%, suggesting that when a usable utterance reached the conversational pipeline, the system generally maintained context and responded appropriately to the participant's intended direction.

The weakest measures were turn-taking and floor management. These failures were concentrated particularly in P-03, where several participant utterances were cut off or missed completely, including transitions between artworks. This frequently produced a secondary contextual failure: the generated response was coherent relative to the context available to the language model but was no longer relevant to the participant's actual conversational state.

Failure Analysis

The quantitative and conversational evaluations identified where interaction quality degraded, but did not by themselves distinguish between failures of the underlying conversational design and failures of individual runtime components. The dialogue transcripts, telemetry and participant interviews were therefore examined together to identify recurring failure families and trace their likely origin.

Eight failure families were observed. Some represent limitations of the current architecture, while others arise from the technical behaviour of particular speech, streaming or playback components. A third group are mixed failures, in which a component-level problem is amplified by the surrounding architecture.

Observed failure family	Evidence from evaluation	Diagnosis	Change type	Improvement indicated
ASR misrecognition	Names, artwork titles and ordinary phrases were sometimes transcribed incorrectly. Contextual inference frequently still recovered the intended artwork or request.	Recognition errors originate technically, while the architecture currently provides limited domain information to recognition.	Mixed	Supply the recogniser with conversational/domain context where supported; maintain an index of likely artwork titles, artists and entities.
Complete or partial loss of user speech	Some utterances appeared to be missed from the effective conversation. Telemetry indicates that at least some were captured and transcribed before the associated turn was subsequently cancelled.	The speech itself was not necessarily lost; an already captured turn could be discarded when continuing speech was treated as a new interaction.	Technical	Harden the turn/cancellation lifecycle so valid user content is not discarded when the user continues after a short pause.
Premature turn finalisation	Utterances intended as a single contribution were sometimes split. Incomplete	Turn-completion detection was occasionally too aggressive and could launch	Technical	Tune silence/finalisation behaviour and allow linguistic or TRP evidence of

	fragments could be marked final after a short silence before the participant had finished speaking.	processing before the conversational turn had genuinely ended.		incompleteness to delay finalisation.
Context propagation after a missed turn	Where an artwork-change utterance was lost or cancelled, later questions could be answered using the previous artwork as the active subject.	Primarily a downstream consequence of lost input, with limited ability to detect that contextual state has become inconsistent with the user.	Mixed	Improve turn reliability first; subsequently explore recovery mechanisms for detecting or repairing stale conversational context.
Response latency and variability	Response delay varied substantially between sessions. P-01 experienced comparatively short delays, whereas P-03 frequently experienced much longer waits.	The modular pipeline accumulates latency across multiple stages, while cloud/model timing also exhibits run-to-run variability.	Mixed	Continue reducing context-resolution and first-token latency and improve tail-latency reliability rather than optimising only median performance.
Lack of visual/situated grounding	Participants sometimes asked about visible features absent from the retrieved textual artwork description.	Docent currently knows artworks primarily through textual records rather than visual representations.	Architectural	Enrich the retrieval corpus with multimodal/image-derived representations of each artwork.
Insufficient guide initiative and knowledge flexibility	Participants wanted more proactive follow-up, connections between works and recommendations based on earlier interests.	Current retrieval and prompting prioritise grounded local answering over broader guide-like initiative and flexible use of available knowledge.	Architectural	Develop richer recommendation/index retrieval and clearer policies for when to use retrieved evidence, model knowledge or other trusted sources.
Audio playback disruption	Audio-buffer underruns occurred during some responses, particularly where synthesis throughput approached the rate at which audio was being consumed.	Audio synthesis/playback occasionally lacked sufficient buffering to absorb variations in generation rate.	Technical	Introduce more resilient or adaptive pre-buffering and monitor synthesis real-time factor.

Turn finalisation and apparently “lost” speech

The second and third failure families are closely related but were retained separately because they produce different observable behaviours. Premature finalisation describes the system incorrectly deciding that the participant has finished speaking. Loss of speech describes the later consequence when that prematurely finalised contribution is cancelled after the participant continues.

The observed sequence can therefore be represented as: brief pause → premature turn finalisation → response processing begins → participant continues → continued speech treated as new turn → previous turn cancelled.

The resulting interaction may appear to the participant as if Docent simply 'did not hear' part of what was said, despite the audio having initially reached the transcription pipeline. This distinction is significant because it directs improvement towards turn-management logic rather than ASR accuracy alone.

The failure analysis therefore suggests that the prototype's remaining limitations are not concentrated in a single component. The semantic dialogue behaviour was generally successful once a usable user turn entered the system, while the most consequential breakdowns occurred either before that point, through turn-processing failures, or where the architecture lacked information required by the user's situated question. These findings provide the basis for the subsequent discussion of future work and possible architectural extensions.

Reference

Walker, M. A., Litman, D. J., Kamm, C. A., & Abella, A. (1998). Evaluating spoken dialogue agents with PARADISE: Two case studies.